



# UNIVERSITY OF MINES AND TECHNOLOGY, TARKWA

## FIRST SEMESTER EXAMINATION, APRIL 2024

**COURSE NO:** MC/EL/GM/GL/ES/CE 361

**COURSE NAME:** PROBABILITY AND STATISTICS

**CLASS:** MC/EL/GM/GL/ES/CE III

**TIME:** 3 HOURS

Name: \_\_\_\_\_

Index Number: \_\_\_\_\_

### SECTION A

**INSTRUCTION:** *ANSWER ALL QUESTIONS BY CIRCLING THE RIGHT ANSWER ON THE QUESTION PAPER.*

1. Which of the following mentioned standard Probability density functions is applicable to continuous Random Variables?
  - a. Binomial Distribution
  - b. Exponential Distribution
  - c. Poisson Distribution
  - d. Bernoulli Distribution
  
2. What would be the probability of an event 'G' if H denotes its complement, according to the axioms of probability?
  - a.  $P(G) = \frac{1}{P(H)}$
  - b.  $P(G) = 1 - P(H)$
  - c.  $P(G) = 1 + P(H)$
  - d.  $P(G) = P(H)$
  
3. The expected value of continue random variable "x" is given by
  - a.  $P(x)$
  - b.  $\sum x P(x)$
  - c.  $\int x P(x)$
  - d. No value such as expected value
  
4. A stick of unit length is broken randomly into three pieces. (Specifically, two points a; b are chosen at random on the stick and it is cut at these two points.) What is the probability that the three stick pieces can be formed into a triangle?
  - a.  $\frac{1}{4}$
  - b.  $\frac{1}{2}$
  - c. 1
  - d.  $\frac{1}{8}$

5. A variable that can assume any value between two given points is called
- Continuous random variable
  - Discrete random variable
  - Irregular random variable
  - Random variable
6. If  $A$  and  $B$  are two mutually exclusive events with  $P(a) > 0$  and  $P(b) > 0$  then it implies they are also independent
- True
  - False
7. Let  $A$  and  $B$  be two events such that  $P(A) = 1/5$  While  $P(A \text{ or } B) = 1/2$ . For what values of  $P$  are  $A$  and  $B$  independent?
- $1/4$
  - $1/2$
  - 0, since mutually exclusive
  - $5/18$
8. If  $A$  and  $B$  are two events such that  $P(a) = 0.2$ ,  $P(b) = 0.6$  and  $P\left(\frac{A}{B}\right) = 0.2$  then the value of  $P(A/\sim B)$  is
- 0.2
  - 0.5
  - 0.8
  - $1/3$
9. Two sisters maintain that they can communicate telepathically. To test this assertion, you place the sisters in separate rooms and show sister A, a series of cards. Each card is equally likely to depict either a circle or a star or a square. For each card presented to sister A, sister B writes down 'circle', or 'star' or 'square', depending on what she believes sister A to be looking at. If ten cards are shown, what is the probability that sister B correctly matches at least one?
- 0.9910
  - 0.9827
  - 0.5000
  - 0.9100
10. A statement made about a population for testing purpose is called?
- Statistic
  - Hypothesis
  - Level of significance
  - Test-statistics

11. A process for putting chocolate chips into cookies is random and the number of choc chips in a cookie has a Poisson distribution with mean  $\lambda$ . Find an expression for the probability that a cookie contains less than 3 choc chips.

|                                                                               |                                                                                 |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| a. $P(X < 3) = e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2}\right)$   | c. $P(X > 3) = e^{-\lambda} \left(1 + \lambda + \frac{\lambda^4}{4}\right)$     |
| b. $P(X < 3) = e^{-\lambda} \left(1 + \lambda^3 + \frac{\lambda^3}{3}\right)$ | d. $P(X \geq 3) = e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2!}\right)$ |

12. A manufacturer produces piston rings for an automobile engine. It is known that ring diameter is  $\sigma = 0.001$  millimeter. A random sample of 15 rings has a mean diameter of  $\bar{x} = 74.036$  millimeters. Construct a 99 % two-sided confidence interval on the mean piston ring diameter.

|                                    |                                    |
|------------------------------------|------------------------------------|
| a. $74.0355 \leq \mu \leq 74.0367$ | c. $60.0011 \leq \mu \leq 61.2231$ |
| b. $72.0911 \leq \mu \leq 70.0111$ | d. None of the above               |

13. Suppose  $n$  balls are distributed at random into  $r$  boxes. Find the probability that there are exactly  $k$  balls in the first  $r_1 (< r)$  boxes.

|                                                                                                 |                                                                                                 |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| a. $\left(\frac{n}{k}\right) \left(\frac{r_1}{k}\right)^k \left(\frac{r_1}{r}\right)^{n-k}$     | c. $\left(\frac{n}{k}\right) \left(1 - \frac{r_1}{k}\right) \left(\frac{r_1}{r}\right)^{n-k}$   |
| b. $\left(\frac{n}{k}\right) \left(\frac{r_1}{r}\right)^k \left(1 - \frac{r_1}{r}\right)^{n-k}$ | d. $\left(1 - \frac{n}{k}\right)^k \left(\frac{r_1}{k}\right) \left(\frac{r_1}{r}\right)^{n-k}$ |

14. A regression analysis yields the line

$$\hat{y} = 18 + 0.25x$$

One of the subjects, Mary, has  $x = 40$  and  $y = 32$ . Calculate Mary's residual.

|      |      |      |      |
|------|------|------|------|
| a. 6 | b. 2 | c. 4 | d. 8 |
|------|------|------|------|

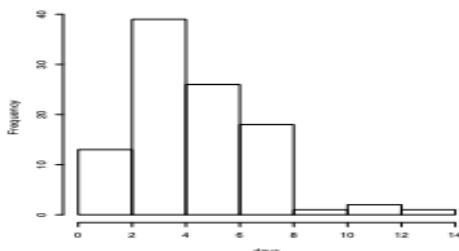
15. During the course of an experiment with a particular brand of tractor engines, it is found that 80% of the tractor engines are damaged in the first experiment. The tractor engines that survived developed significant amount of wear and tear, so that the percentage of engines that survived in any later experiment is half that in the preceding experiment. Find the probability that an engine will survive 4 experiments.

|         |         |         |         |
|---------|---------|---------|---------|
| a. 0.08 | b. 0.05 | c. 0.45 | d. 0.11 |
|---------|---------|---------|---------|

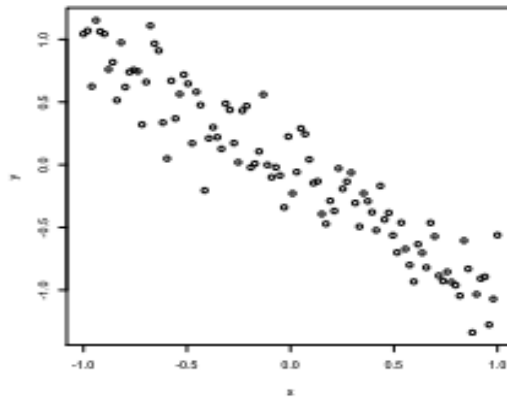
16. Which of the following statistical techniques should be used when you wish to examine the association between two nominal variables?
- Multiple regression analysis
  - Simple correlation coefficient
  - Pearson correlation coefficient
  - Chi-square contingency test
17. Find the number of ways of arranging the letters of the words DANGER, so that no vowel occupies odd place.
- 144.
  - 44
  - 11
  - 124
18. Let  $X$  be a random variable on with probability density function (pdf)  $f(x) = cx^2$  defined on  $0 < x < 1$ .
- 3
  - 1
  - c
  - $1/3$
19. If two events are independent, then
- They must be mutually exclusive
  - The sum of their probabilities must be equal to one
  - Their intersection must be zero
  - None of the above
20. If a hypothesis is rejected at the 5% level of significance, it
- Will always be rejected at the 1% level
  - Will always be accepted at the 1% level
  - Will never be tested at the 1% level
  - May be rejected or not rejected at the 1% level
21. If the probability that a bomb dropped from a place will strike the target is 60% and if 10 bombs are dropped, find mean and variance?
- 6, 0.24
  - 0.4, 0.16
  - 0.6, 0.24
  - 4, 1.6
22. If  $P(1) = P(3)$  in Poisson's distribution, what is the mean?
- $\sqrt{2}$
  - $\sqrt{3}$
  - $\sqrt{6}$
  - $\sqrt{7}$

23. What is the mean and variance for standard normal distribution?
- Mean is 0 and variance is 0
  - Mean is 0 and variance  $\infty$
  - Mean is  $\infty$  and variance is 0
  - Mean is 0 and variance is 1
24. Given the random variables  $X, Y$  with joint function with density  $f(x, y) = \frac{2}{3}(x + y)$  on interval  $0 < x < 1$  and  $0 < y < 2$ . What is the relationship between  $X$  and  $Y$ ?
- Negative relationship
  - Inconclusive
  - Positive relationship
  - None of the above
25. Appropriate graphical summary of the distribution of a categorical variable
- Bar chart
  - Box plot
  - Residual plot
  - Scatter plot
26. What does probability sampling allow us to do?
- Make inferences about population parameter
  - Remove sampling variability
  - Assess cause and effect relationship
  - Exactly represents the population
27. If  $E(x) = 2$  and  $E(z) = 4$ , then  $E(z - x) = ?$
- 0
  - Insufficient data
  - 2
  - 1
28. A national random sample of 20 BECE scores from 2023 is listed below. Calculate the sample mean and standard deviation.
- 29, 26, 13, 23, 23, 25, 17, 22, 17, 19, 12, 26, 30, 30, 18, 14, 12, 26, 17, 18
- 20.50; 5.79
  - 20.50; 5.94
  - 20.85; 5.79
  - 20.85; 5.94
29. Provided that the BECE is reasonably normally distributed with a mean of 18 and standard deviation of 6, determine the proportion of students with a 33 or higher.
- 0.0062
  - 0.0109
  - 0.0124
  - 0.0217
30. The data in question 28 appears to be reasonably symmetric
- True
  - False

31. Select the order of sampling schemes from best to worst.
- |                                           |                                           |
|-------------------------------------------|-------------------------------------------|
| a. simple random, stratified, convenience | c. stratified, simple random, convenience |
| b. simple random, convenience, stratified | d. stratified, convenience, simple random |



32. The histogram above represents the lifespan of a random sample of a particular type of insect. Determine the relationship between the mean and median.
- |                          |                    |
|--------------------------|--------------------|
| a. $mean = median$       | c. $mean < median$ |
| b. $mean \approx median$ | d. $mean > median$ |
33. When the correlation coefficient,  $r$ , is close to one:
- |                                                                    |                                                                                  |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------|
| a. there is no relationship between the two variables              | c. it is impossible to tell if there is a relationship between the two variables |
| b. there is a strong linear relationship between the two variables | d. the slope of the regression line will be close to one                         |
34. The distribution of heights of American women aged 18 to 24 is approximately normally distributed with a mean of 65.5 inches and standard deviation of 2.5 inches. Calculate the z-score for a woman six feet tall.
- |          |         |          |         |
|----------|---------|----------|---------|
| a. 2.60. | b. 4.11 | c. 1.04. | d. 1.33 |
|----------|---------|----------|---------|
35. In 1923, Babe Ruth had 522 at bats with 205 hits. Assuming that the binomial distribution can be appropriately applied, find the expected number of hits in 529 at bats
- |        |        |        |        |
|--------|--------|--------|--------|
| a. 321 | b. 321 | c. 186 | d. 208 |
|--------|--------|--------|--------|



36. The data in the scatterplot above would have a correlation coefficient that is close to:
- 1.0.
  - 0.5
  - +1.0
  - 0.5
37. The intercept in linear regression represents:
- the strength of the relationship between x and y
  - the expected x value when y is zero
  - the expected y value when x is zero
  - a population parameter
38. Mean of a constant 'a' is
- 0
  - a
  - $a/2$
  - 1
39. Suppose that a game is played with a single die assumed fair. In this game a player wins \$20 if a 2 turns up, \$40 if a 4 turns up; loses \$30 if a 6 turns up; while he neither wins nor loses if any other face turns up. The expected sum of money won is
- 1
  - 0
  - 6
  - 5
40. Find the probability that a person tossing three coins will get either all heads or all tails for the second time in the fifth toss?
- $\frac{1}{156}$
  - $\frac{1}{4}$
  - $\frac{1}{8}$
  - $\frac{27}{256}$
41. In a certain theodolite manufacturing process, it is known that on the average, 1 in every 100 is defective. If we are interested in computing the probability that the fifth item inspected is the first defective theodolite found? Which distribution will be applied in computing for the probability
- Bernoulli distribution
  - Geometric distribution

b. Binomial distribution

d. Poisson distribution

## SECTION B

**Carefully read each question and circle the CORRECT ANSWERS ONLY. Show all workings in the Answer Booklet provided. Wrong answers with clear and correct workings would attract some point(s)**

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Use the following information to answer Questions 42 to 45.

**PROBLEM 1:** A contractor is required by a county planning department to submit one, two, three, four, five or six forms (depending on the nature of the project) in applying for a building permit. Let  $Y = \text{the number of forms required of the next applicant}$ . The probability that  $y$  forms are required is known to be proportional to  $y$  that is  $p(y) = ky$  for  $y = 1, \dots, 6$ .

42. What is the value of  $k$  ?

- a.  $k = 1/22$       b.  $k = 2/22$       c.  $k = 3/22$       d.  $k = 6/22$

43. What is the probability that at most three forms are required?

- a.  $k = 3/7$       b.  $k = 1/7$       c.  $k = 2/7$       d.  $k = 1/6$

44. What is the probability that between two and four forms (inclusive) are required?

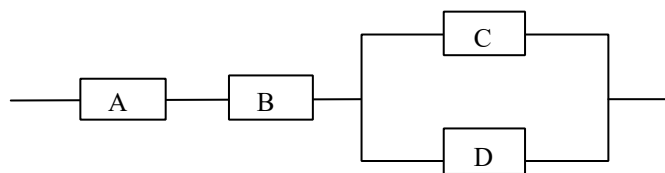
- a.  $k = 3/7$       b.  $k = 1/7$       c.  $k = 2/7$       d.  $k = 1/6$

45. Could  $P(Y) = y^2/50$  for  $y = 1, \dots, 6$  be the probability mass function (pmf) of  $Y$  ?

- a. For  $c = 91$  the function  $P(Y)$  becomes a pmf  
b. Since  $91/50$  does not equal 1, the function  $P(Y)$  cannot be the pmf for  $Y$   
c. The function may or may not be a PMF since we do not know the value of  $Y$ .  
d.  $P(Y)$  is a PDF

Use the following information to answer Questions 46 to 48.

**PROBLEM 2:** An electrical system consists of four components as illustrated in the figure below.



The system works if components  $A$  and  $B$  work and either of the components  $C$  or  $D$  work. It is known that the components work independently and that  $P(A \text{ works}) = P(B \text{ works}) = 0.9$  and  $P(C \text{ works}) = P(D \text{ works}) = 0.8$ .

46. What is the probability that an entire system works?
- a. 0.76
  - b. 0.77
  - c. 0.78
  - d. 0.79
47. What is the probability that the component C does not work, given that the entire system works?
- a. 0.66
  - b. 0.76
  - c. 0.17
  - d. 0.18
48. What is the probability distribution of the system's time to failure?
- a. Weibull distribution
  - b. Normal distribution
  - c. Exponential distribution
  - d. Binomial distribution

Use the following information to answer Questions 49 to 50.

**PROBLEM 3:** A factory has a machine that dispenses 80 mL of fluid in a bottle. An employee believes the average amount of fluid is not 80 mL. Using 40 samples, he measures the average amount dispensed by the machine to be 78 mL with a standard deviation of 2.5.

49. What is the null and research hypothesis
- a.  $H_o: \mu \neq 80$   
 $H_a: \mu = 80$
  - b.  $H_o: \mu \neq 80$   
 $H_o: \mu \geq 80$
  - c.  $H_o: \mu = 80$   
 $H_o: \mu \neq 80$
  - d.  $H_o: \mu \geq 80$   
 $H_o: \mu < 80$
50. At a 95% confidence level, is there enough evidence to support the idea that the machine is not working properly based on the zcore?
- a. It is believe that machine is working at a 95% confidence level
  - b. It is believed that the machine is not working properly at a 95% confidence level
  - c. It is believed that the machine is working properly
  - d. It is believed that the machine does not work properly at 0.05 critical level